Structure and Structure Dependence

Sentence structure • hierarchy • constituency • constituency tests

What is a sentence?

A sentence is a grammatically organized linguistic expression that can function as an independent unit of communication.

Grammar determines how words combine into phrases and clauses and how those units combine into larger structures.

A sentence is therefore not just a string of words; it has internal organization.

Example: The students read the article.

Words: The | students | read | the | article

Structure: [The students] [read [the article]]

Requirements of a well-formed sentence

Lexical requirement: words must be appropriate for the intended construction.

Structural requirement: words must combine in permitted hierarchical configurations.

Category requirement: phrases generally have identifiable syntactic categories (NP/DP, VP, PP, etc.).

Grammatical relation requirement: elements can bear relations such as subject and object.

Agreement requirement: where the language requires it, features such as person and number must be compatible.

Semantic/pragmatic interpretability often contributes to acceptability, but grammaticality and meaningfulness are not identical.

“Language is hierarchical” — what does this mean?

Words are grouped into constituents; constituents can themselves contain smaller constituents.

A larger constituent can therefore contain another constituent: a phrase can contain a phrase.

Example: [The very old [wooden house]]

“wooden house” is inside the larger noun phrase “the very old wooden house”.

Hierarchy captures relationships of dominance/containment that linear order cannot capture.

Two words can be adjacent without forming a constituent, while non-adjacent material may sometimes form a constituent in other constructions.

Structure dependence

Structure dependence means that grammatical operations are sensitive to hierarchical structure, not simply to the linear position of words.

A rule cannot generally be stated as “take the third word” or “move the first noun.”

Instead, it refers to structurally defined units such as subject NP, VP, auxiliary, complement, or clause.

This is especially clear in questions, negation, agreement, movement, and binding.

What is a constituent?

A constituent is a syntactic unit whose members form a structural group.

Constituents can be single words or multiword phrases.

Examples: [the students] (NP), [very quickly] (AdvP), [in the library] (PP), [read the book] (VP).

Constituency is a structural property—not simply a matter of words being next to each other.

Constituency tests are diagnostics: they provide evidence for or against treating a sequence as a constituent.

Why do we need constituency tests?

Intuition alone can be unreliable, especially in long or ambiguous sentences.

Bracketing every possible phrase by inspection can lead to competing analyses.

Constituency tests provide observable evidence from grammatical behavior.

No single test is perfect in every context.

Strong analysis usually combines several diagnostics and considers the construction being tested.

Test 1 — Replacement (substitution) test

If a sequence can be replaced by a single pro-form or another syntactically appropriate element, this supports treating the sequence as a constituent.

Common English replacements include pronouns, do so, there, then, and other pro-forms, depending on category and construction.

The replacement must preserve the relevant grammatical function.

Example: The students read the article. → They read the article.

“The students” behaves as a unit and can be replaced by “they”.

Replacement test — more examples

NP replacement
I met [the new professor]. → I met [her].

PP replacement
She lives [in New Delhi]. → She lives [there].

VP replacement
Ravi [read the paper], and Meera [did so] too.

The substitute must be compatible with the category/function of the original phrase.

Replacement by itself does not tell us every detail of internal structure.

Replacement test — limitation

Not every constituent has a convenient pro-form in English.

Some pro-forms are construction-specific: “do so” is mainly associated with verbal material, while “there” can replace certain locative expressions.

Failure to find a natural substitute does not automatically prove that a sequence is not a constituent.

Therefore: positive replacement evidence is useful; negative evidence must be interpreted cautiously.

Test 2 — Movement test

If a sequence can move together as a unit, that supports constituency.

Movement operations include topicalization, fronting, and certain forms of extraposition or scrambling, depending on the language.

Example: The students read the article in the library.

Topicalization: [The article], the students read __ in the library.

The moved material behaves as a single syntactic unit.

Movement test — constituent vs. non-constituent

Constituent

[The old wooden table] collapsed. → [The old wooden table], we moved __ yesterday.

Non-constituent sequence

* [old wooden] the table, we moved __ yesterday.

“The old wooden table” can move as a unit.

“old wooden” is not normally a constituent in this phrase.

The movement test therefore provides structural evidence.

Movement test — important caveat

Not all constituents can freely move in every construction.

Some movement is restricted by information structure, discourse context, lexical properties, or island constraints.

Languages also differ greatly in which movement operations are available.

Thus, inability to move a sequence is not by itself conclusive evidence against constituency.

Use movement together with other diagnostics.

Test 3 — It-cleft

An English it-cleft has the general form: It + BE + focused constituent + relative clause.

Example: It was [the red book] that Maya bought.

The focused phrase occupies a position associated with the information-structural focus of the cleft.

If a sequence fits naturally into the focused constituent position, this supports its constituency.

It-clefts are especially useful for testing NPs, PPs, and other phrases.

It-cleft examples

NP

Maya bought [the red book]. → It was [the red book] that Maya bought.

PP

Maya met him [in the library]. → It was [in the library] that Maya met him.

Adverbial

Maya left [very quickly]. → It was [very quickly] that Maya left.

The clefted sequence is interpreted as one focused unit.

Naturalness can vary by context, so cleftability should be treated as a diagnostic rather than an absolute rule.

Test 4 — Answers to questions

A sequence that can naturally constitute a complete answer to a wh-question often corresponds to a constituent.

Question: What did Maya buy?

Answer: [the red book].

The answer identifies one structural unit corresponding to the information requested by “what”.

This test connects constituency with both syntax and information structure.

Question–answer test: examples

Who?
Who opened the door? → [The teacher].

What?
What did the teacher open? → [the door].

Where?
Where did the teacher open it? → [in the classroom].

When?
When did the teacher arrive? → [after lunch].

Question–answer test — limitation

Answers can be fragments, so their syntax may be less transparent than a full sentence.

Conversational context can license short answers that would not form constituents in the same way under every analysis.

Some answers are pragmatically appropriate even when they do not map neatly onto a single phrase.

Therefore, use question–answer behavior as supporting evidence, not as a standalone proof.

Putting the four tests together

Test	Question it asks	Example evidence
Replacement	Can the sequence be replaced by one pro-form?	the students → they
Movement	Can it move together?	the article → The article, …
It-cleft	Can it be focused as one phrase?	It was the article that…
Question–answer	Can it form a natural answer?	What? → the article

A worked example: “The students read the article.”

Candidate constituents: [The students], [read the article], [the article].

Replacement: [The students] → They; [the article] → it.

Movement: [The article], the students read __.

It-cleft: It was [the article] that the students read.

Question–answer: What did the students read? → [the article].

Converging evidence supports treating these strings as constituents.